Last Time

- **Gradient Flows:**
  - Warm-up: euclidean gradient flows
  - From euclidean to Wasserstein flows
  - The JKO scheme
- **Schrödinger Bridges:**
  - Dynamic OT over path space
  - The Schrödinger problem
  - Connections to entropic OT, Benamou-Brenier
- **Applications in ML**

Last Time

Gradient Flows

- Motivation: often we can't compute entire dynamic OT path, or don't know β
- Instead: **solve problem gradually** by minimizing **functional** $F(\rho)$
- **Gradient Flow** ~ limit of step size $\rightarrow 0$ of grad. descent (Euler), defines continuous curve starting at x_0 that follows $-\nabla f$
- Easy in Euclidean space! Two types of discretization: implicit, explicit.